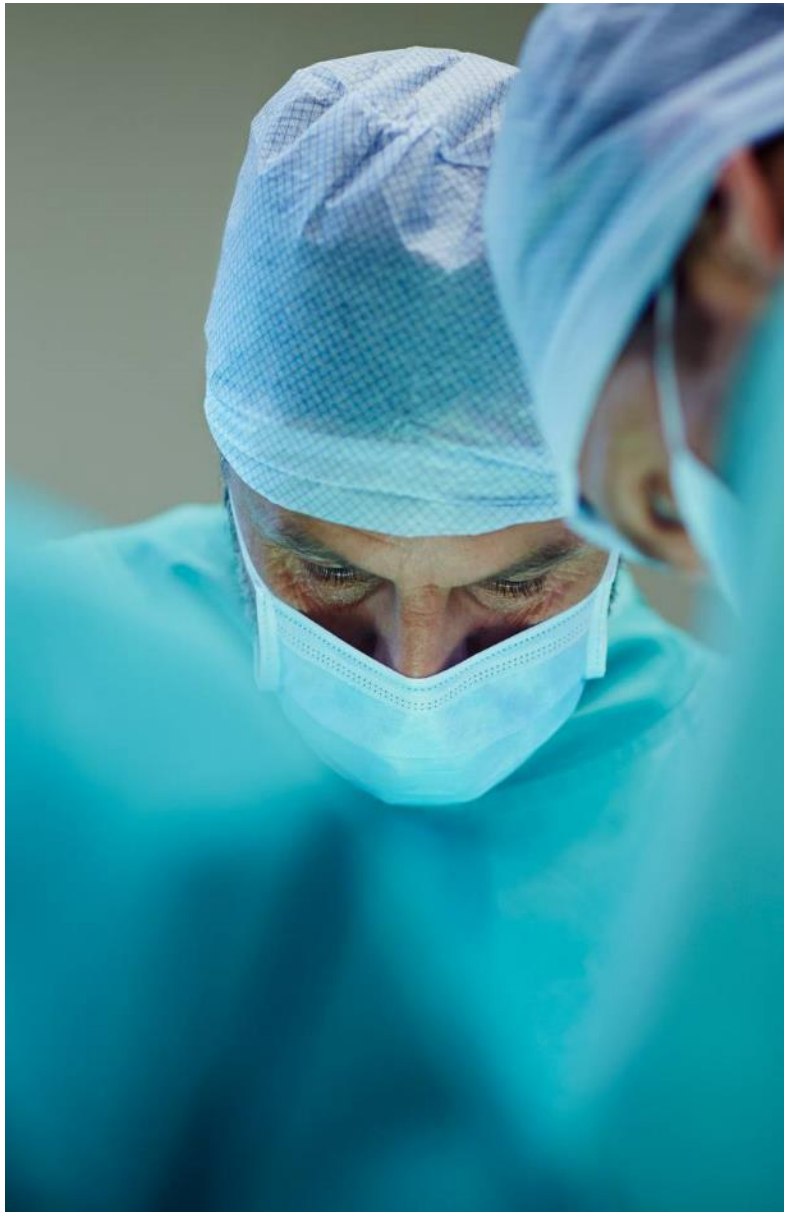
NHS Capacity and Resource Utilisation Report

Exploratory data analysis

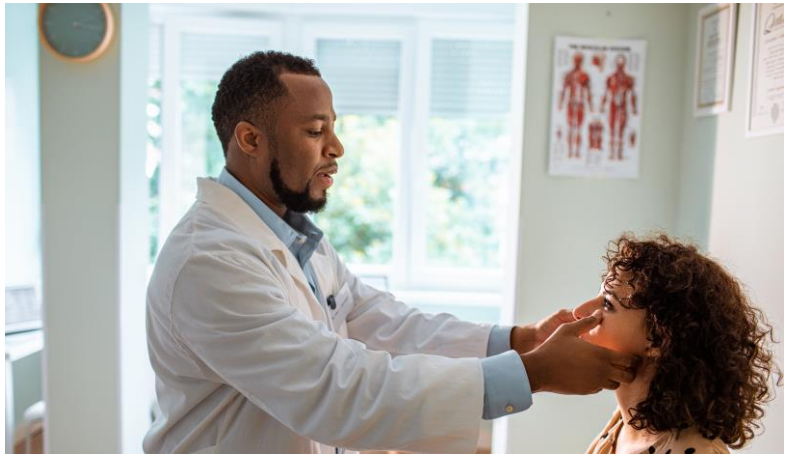
Peter Mackin

Agenda

1. Data Overview
2. Appointment Distribution
3. Capacity
4. Missed Appointments
5. Scenario Analysis
6. Conclusion & Recommendations

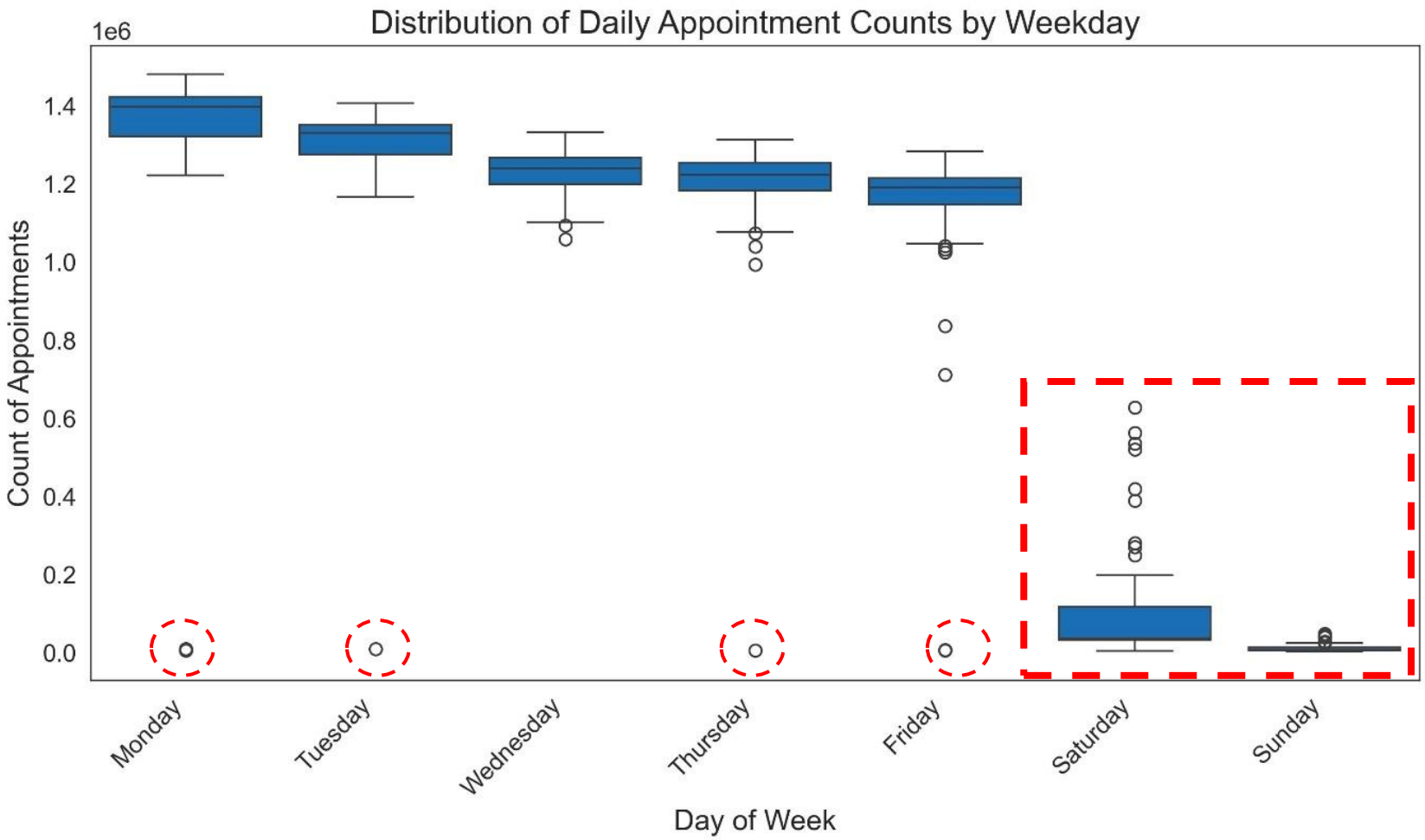


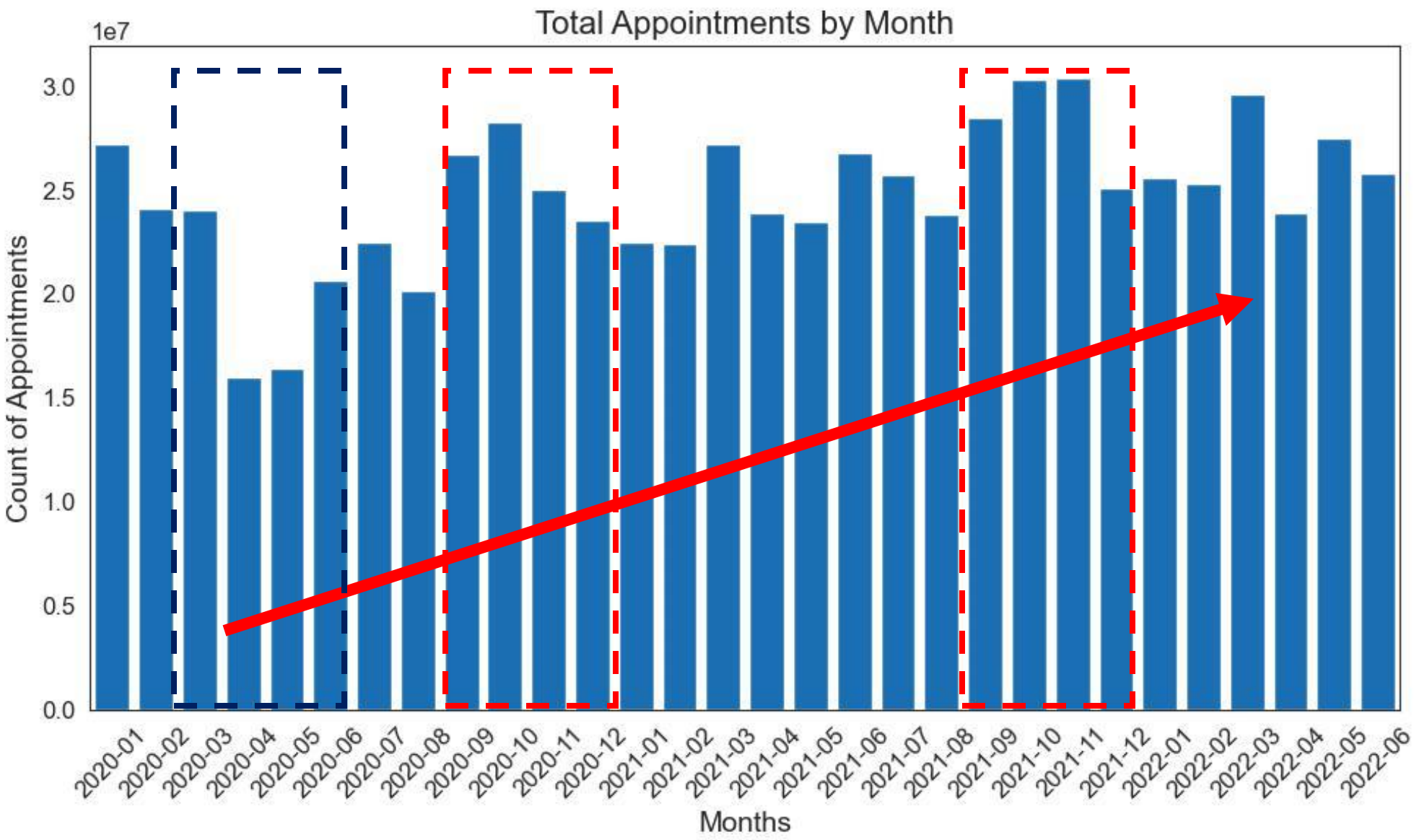
Data Overview

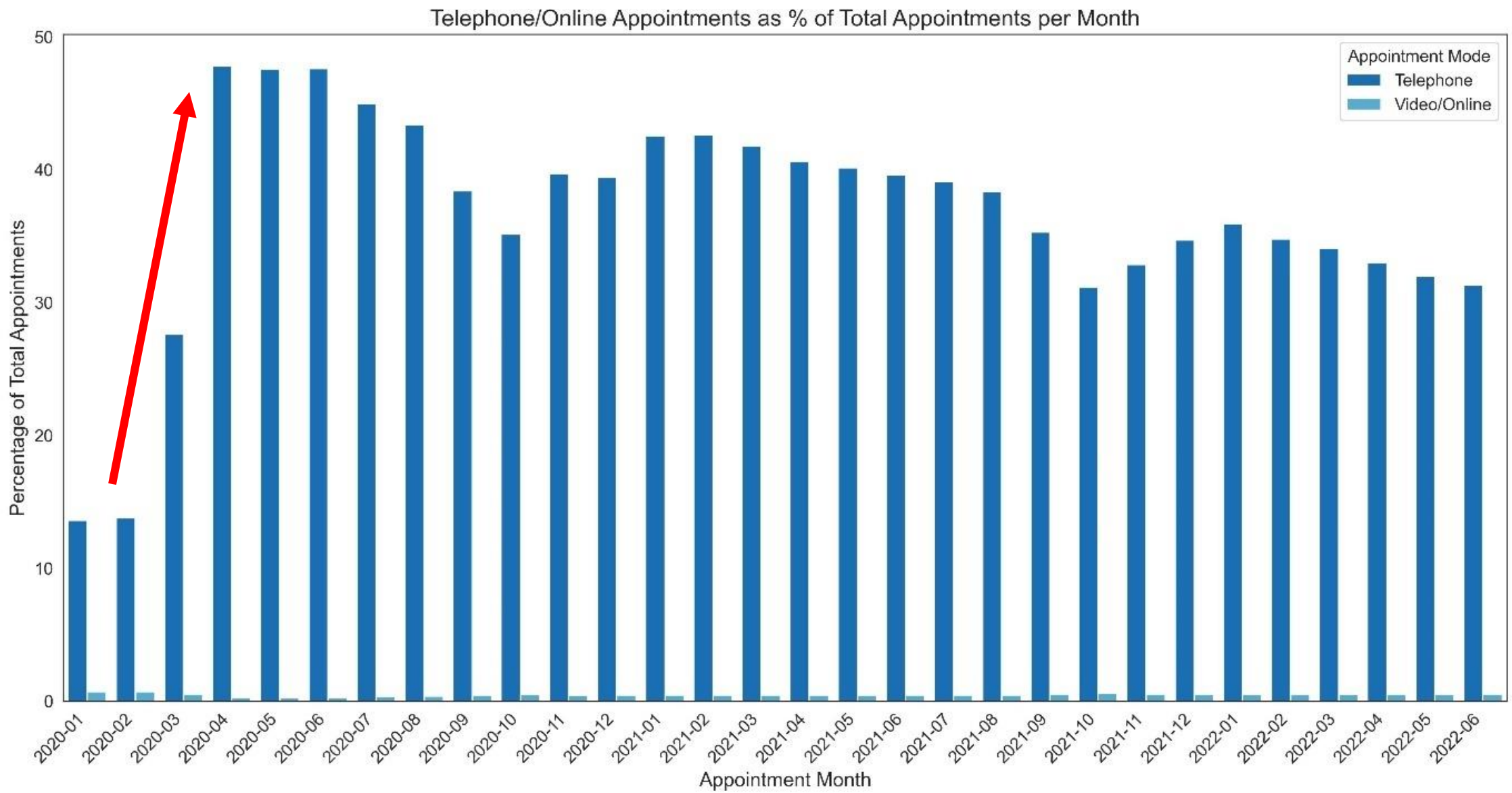


Appointment Distribution

Daily appointment

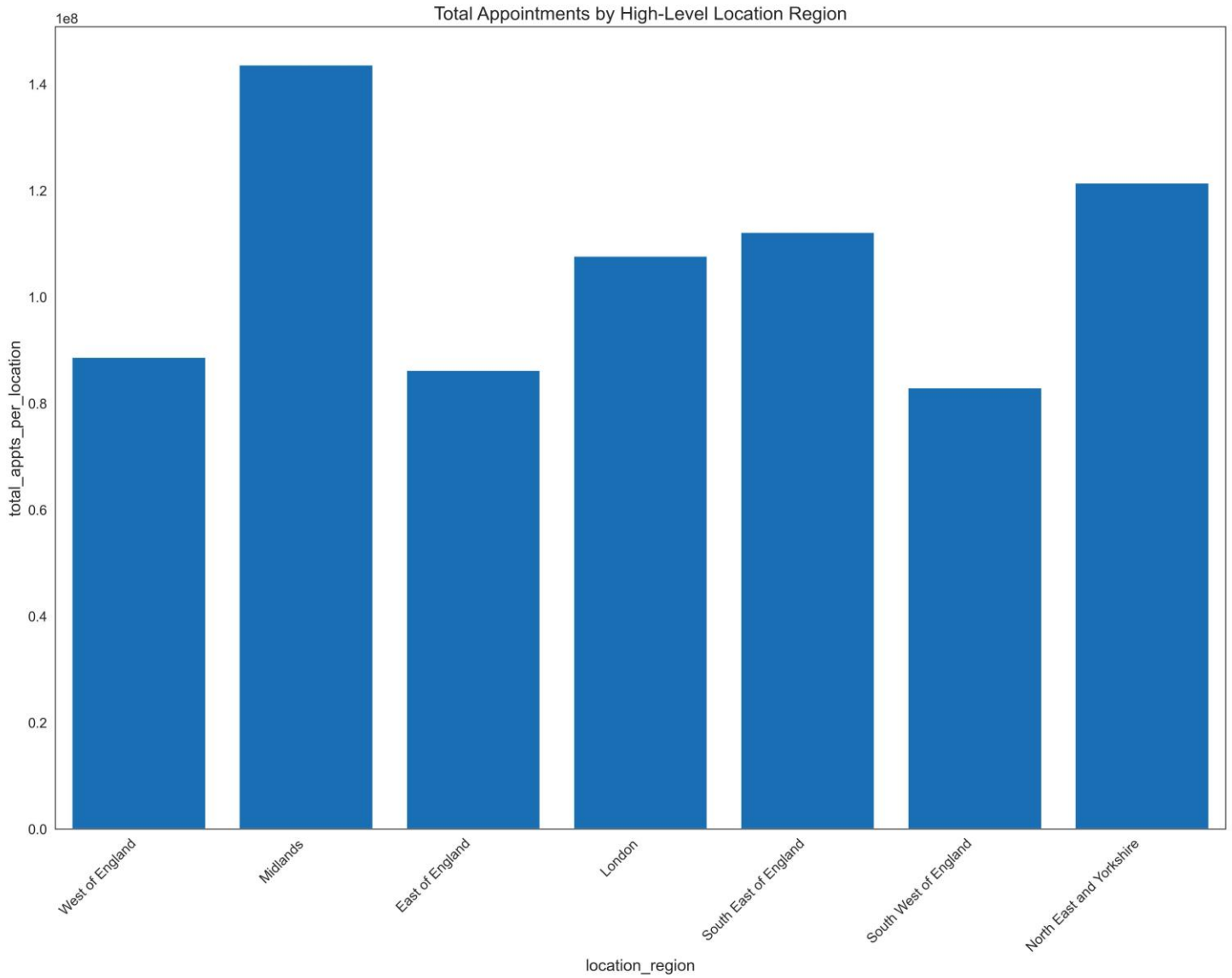


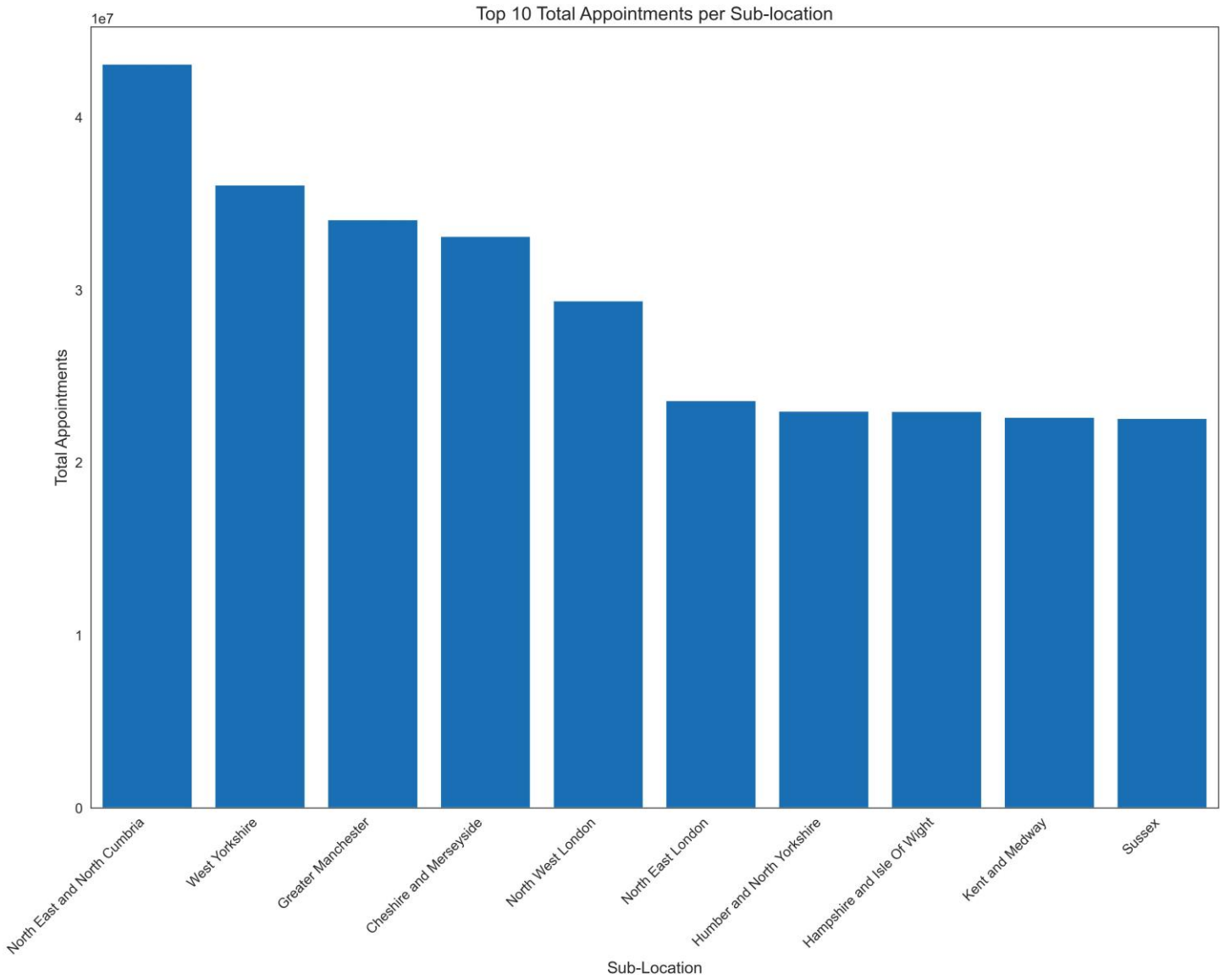

Total Appointments by Month
1e7
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Count of Appointments
2020-01
2020-02
2020-03
2020-04
2020-05
2020-06
2020-07
2020-08
2020-09
2020-10
2020-11
2020-12
2021-01
2021-02
2021-03
2021-04
2021-05
2021-06
2021-07
2021-08
2021-09
2021-10
2021-11
2021-12
2022-01
2022-02
2022-03
2022-04
2022-05
2022-06
Months


Telephone/Online Appointments as % of Total Appointments per Month
Appointment Mode
Telephone
Video/Online
Percentage of Total Appointments
0
10
20
30
40
50
2020-01
2020-02
2020-03
2020-04
2020-05
2020-06
2020-07
2020-08
2020-09
2020-10
2020-11
2020-12
2021-01
2021-02
2021-03
2021-04
2021-05
2021-06
2021-07
2021-08
2021-09
2021-10
2021-11
2021-12
2022-01
2022-02
2022-03
2022-04
2022-05
2022-06
Appointment Month

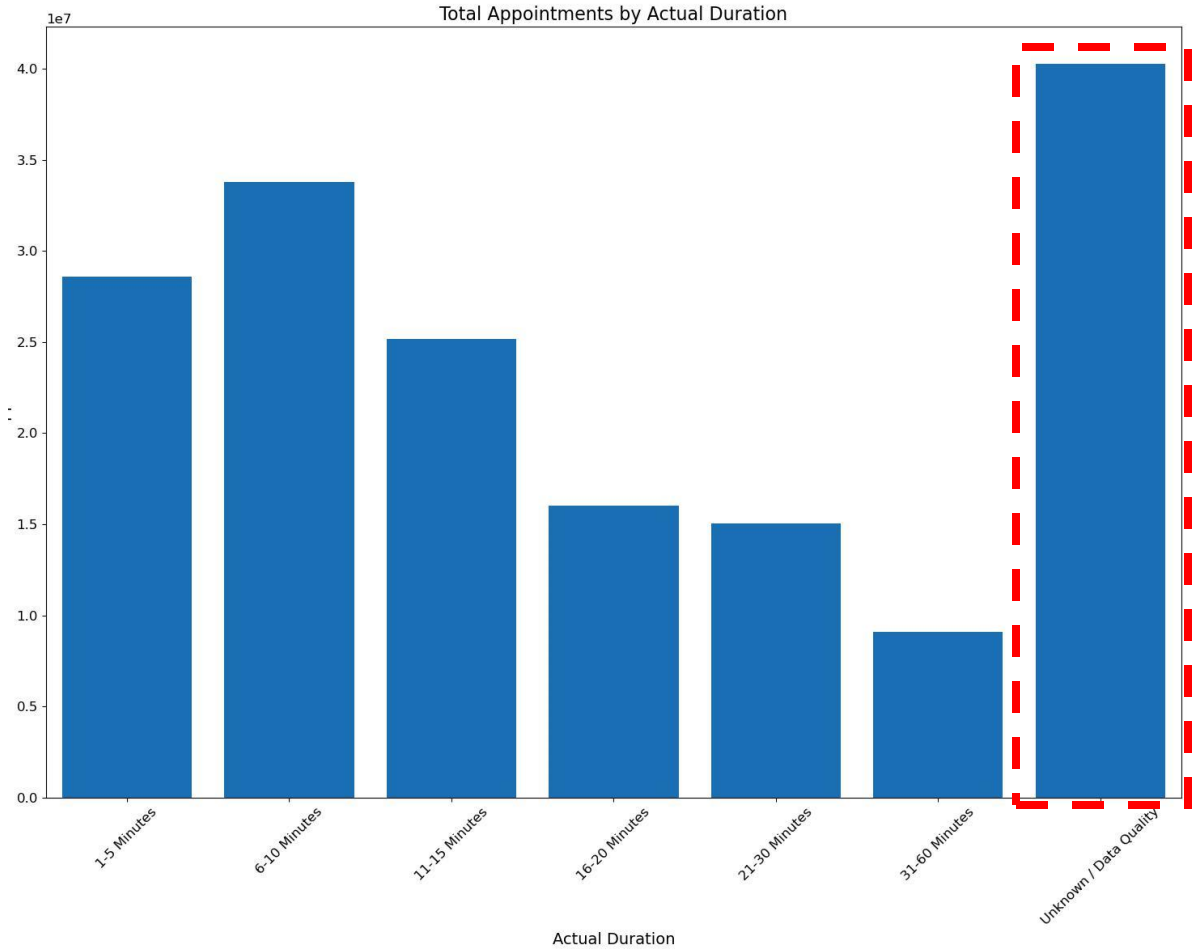
Appointment mode

Mode	Status	% of mode
Face-to-Face	Attended	89%
	DNA	6%
	Unknown	5%
Home Visit	Attended	74%
	DNA	4%
	Unknown	22%
Telephone	Attended	95%
	DNA	2%
	Unknown	3%
Unknown	Attended	88%
	DNA	4%
	Unknown	8%
Video/Online	Attended	91%
	DNA	4%
	Unknown	4%

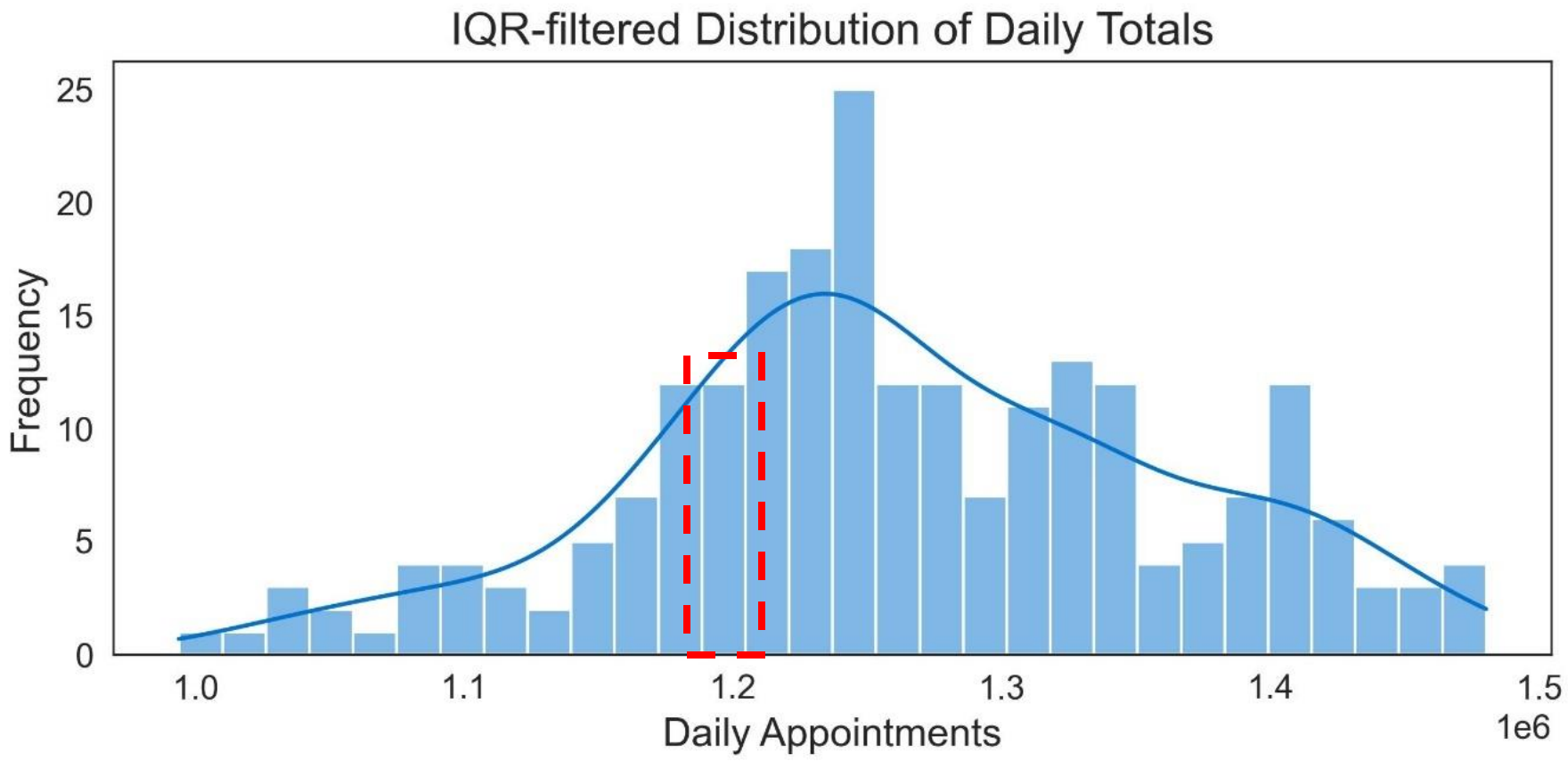

Total Appointments by High-Level Location Region
1e8
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
total_appts_per_location
West of England
Midlands
East of England
London
South East of England
South West of England
North East and Yorkshire
location_region


Top 10 Total Appointments per Sub-location
1e7
4
3
2
1
0
Total Appointments
North East and North Cumbria
West Yorkshire
Greater Manchester
Cheshire and Merseyside
North West London
North East London
Humber and North Yorkshire
Hampshire and Isle Of Wight
Kent and Medway
Sussex
Sub-Location

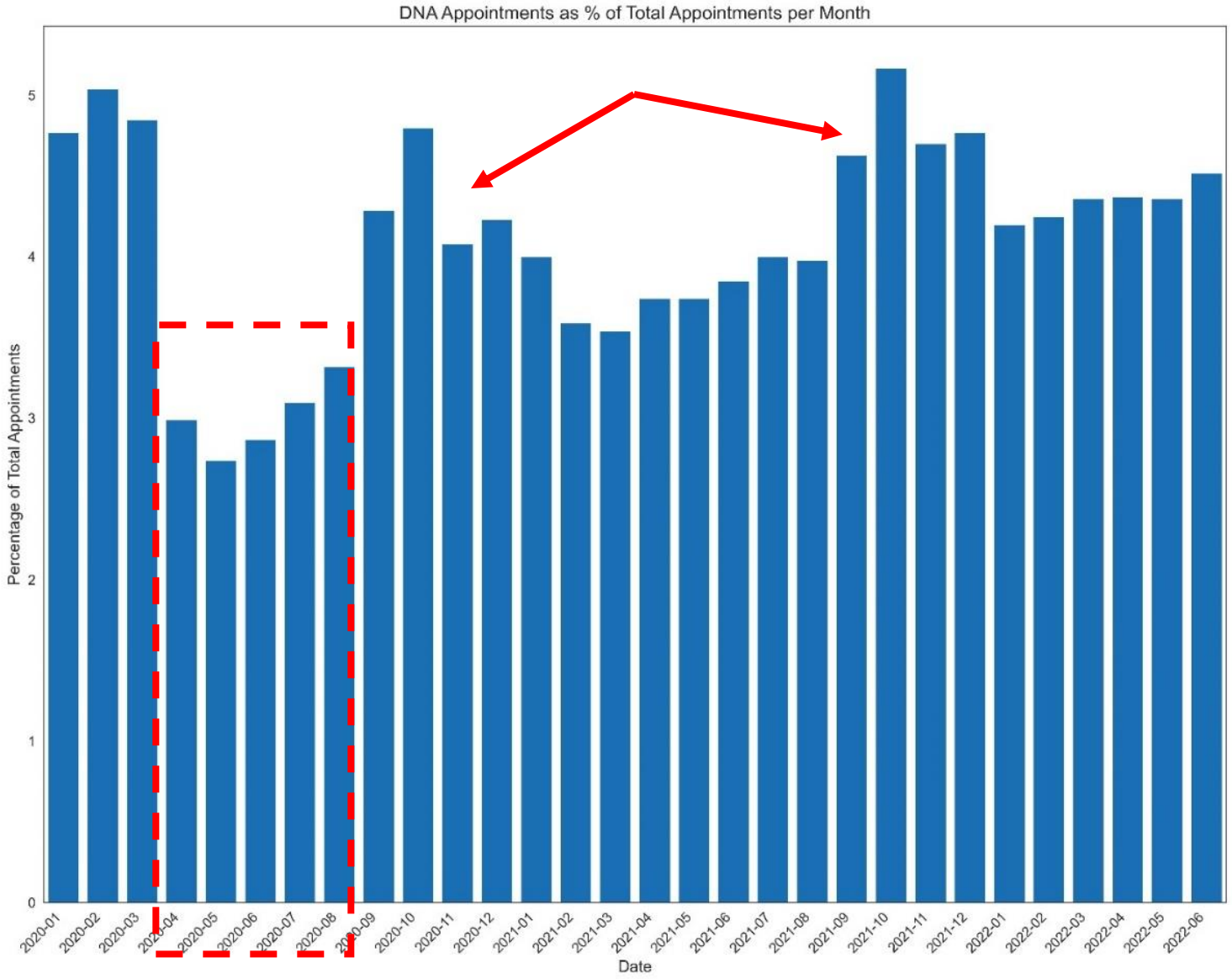
Capacity

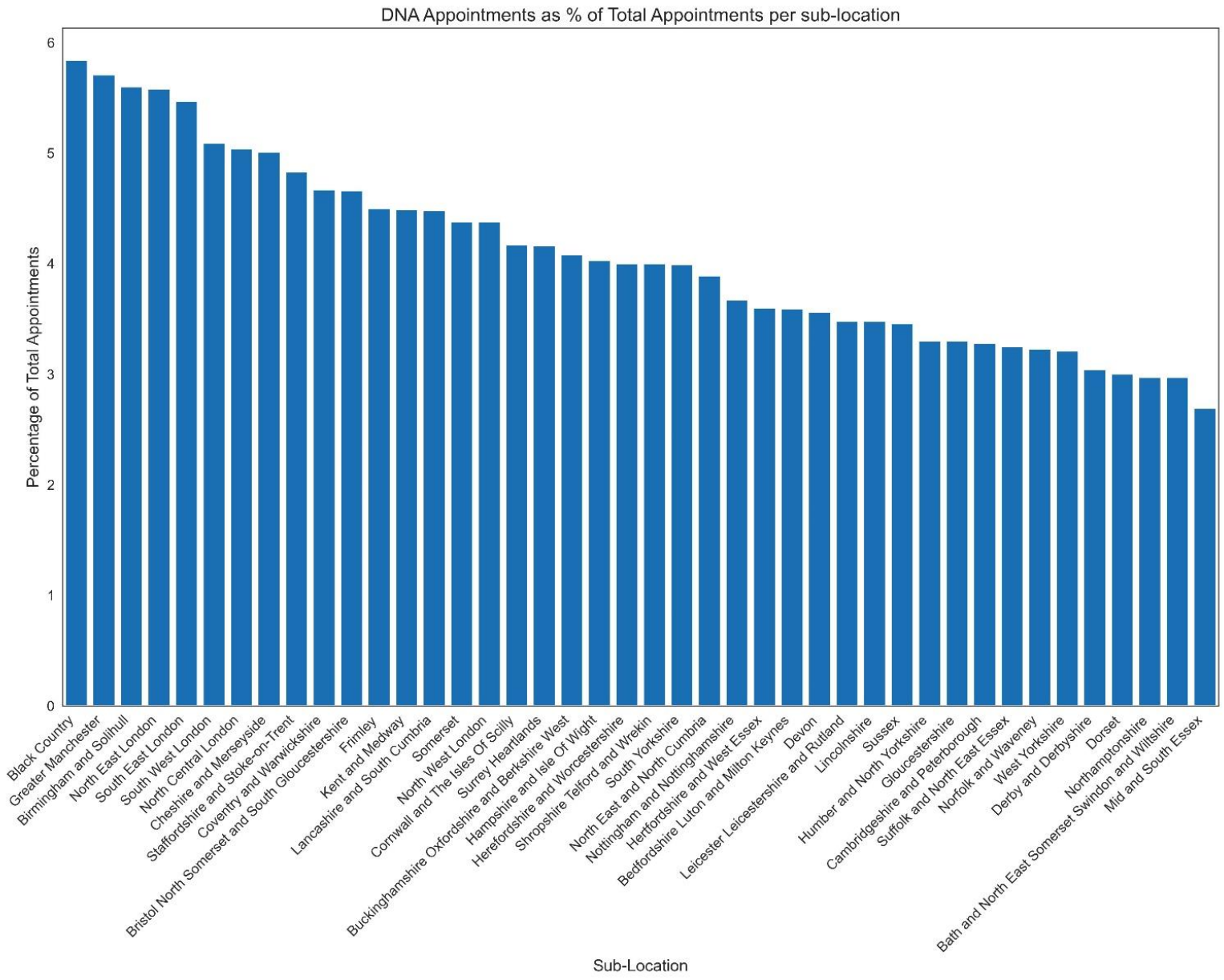

Total Appointments by Actual Duration
1e7
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
1-5 Minutes
6-10 Minutes
11-15 Minutes
16-20 Minutes
21-30 Minutes
31-60 Minutes
Unknown / Data Quality
Actual Duration

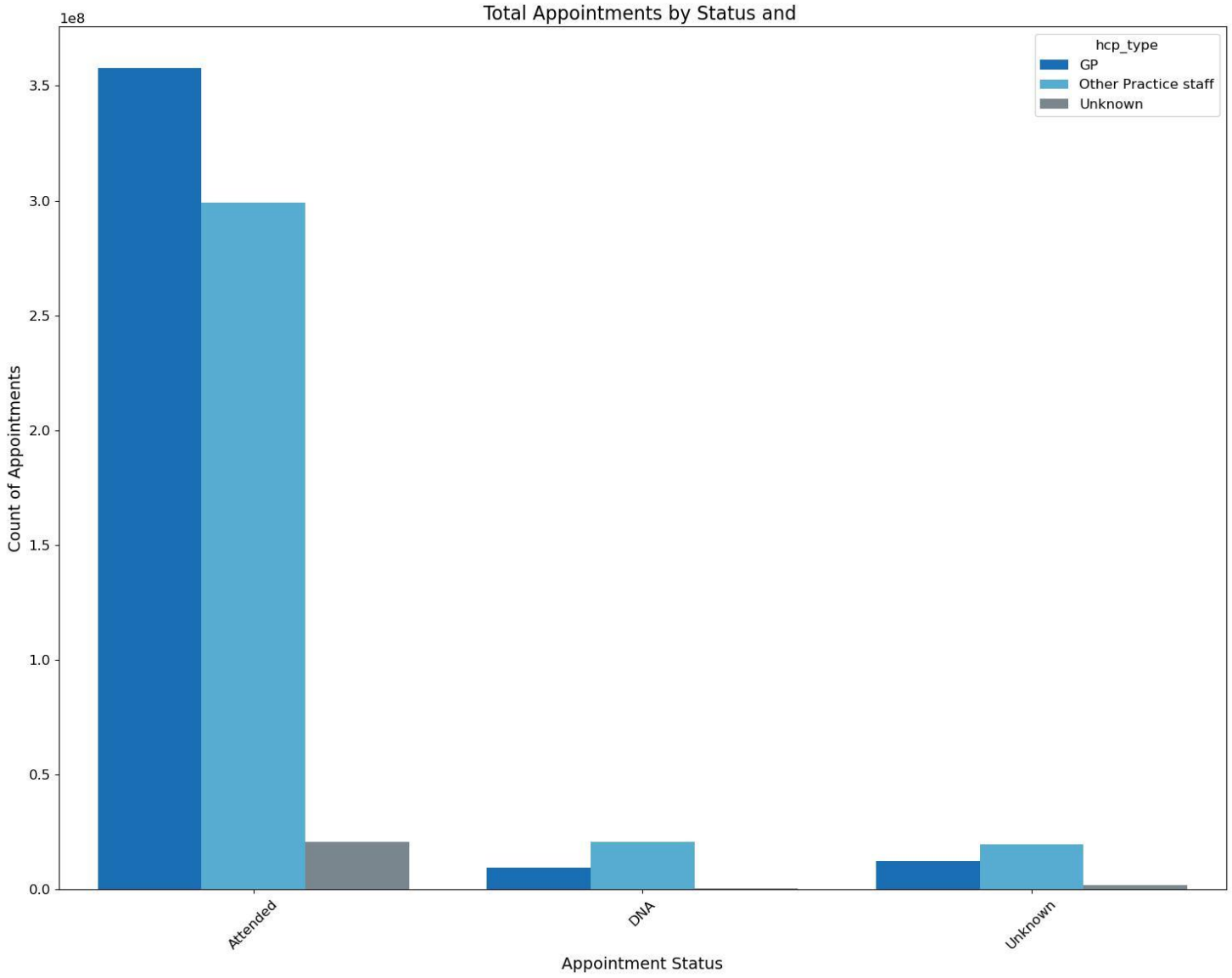
Total daily appointment distribution



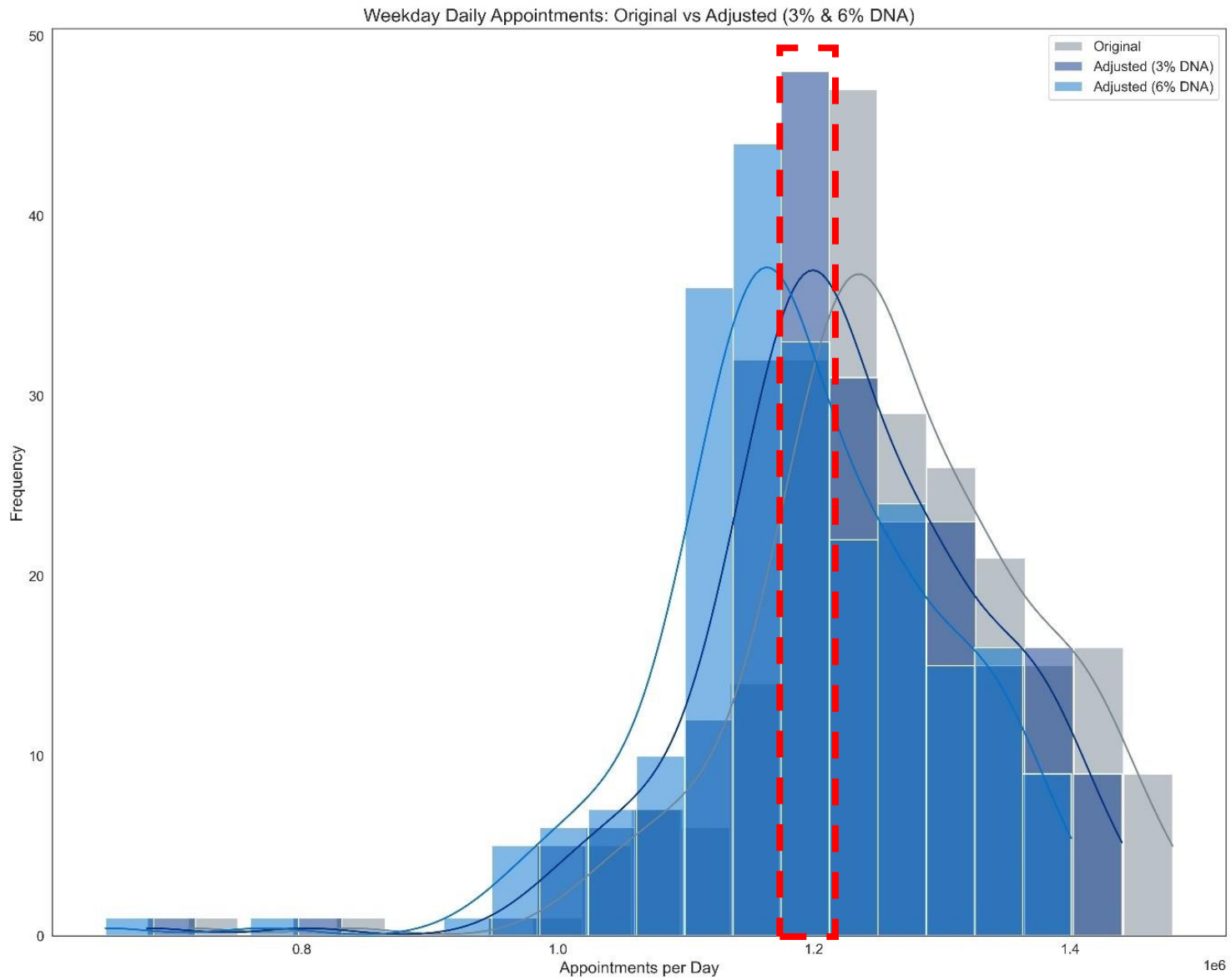
Missed appointments

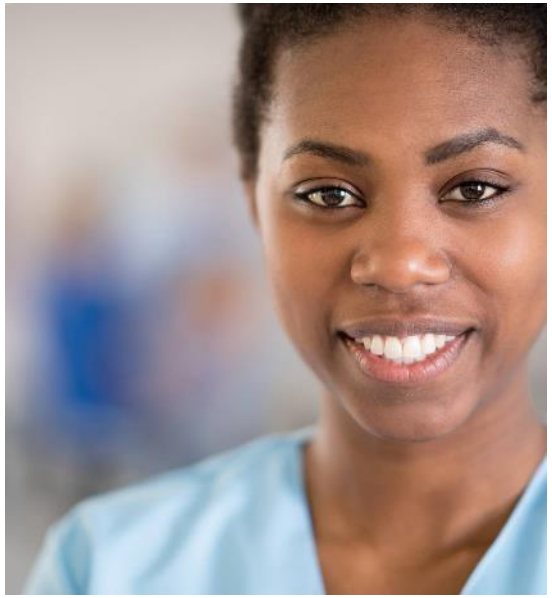
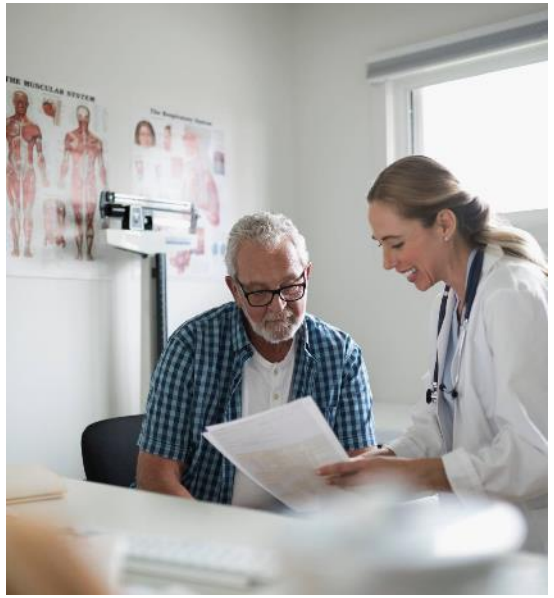

DNA Appointments as % of Total Appointments per Month
Percentage of Total Appointments
0
1
2
3
4
5
2020-01
2020-02
2020-03
2020-04
2020-05
2020-06
2020-07
2020-08
2020-09
2020-10
2020-11
2020-12
2021-01
2021-02
2021-03
2021-04
2021-05
2021-06
2021-07
2021-08
2021-09
2021-10
2021-11
2021-12
2022-01
2022-02
2022-03
2022-04
2022-05
2022-06
Date


DNA Appointments as % of Total Appointments per sub-location
Percentage of Total Appointments
0
1
2
3
4
5
6
Black Country
Greater Manchester
Birmingham and Solihull
North East London
South East London
South West London
North Central London
Cheshire and Merseyside
Staffordshire and Stoke-on-Trent
Coventry and Warwickshire
Bristol North Somerset and South Gloucestershire
Frimley
Kent and Medway
Lancashire and South Cumbria
Somerset
North West London
Cornwall and The Isles Of Scilly
Surrey Heartlands
Buckinghamshire Oxfordshire and Berkshire West
Hampshire and Isle Of Wight
Herefordshire and Worcestershire
Shropshire Telford and Wrekin
South Yorkshire
North East and North Cumbria
Nottingham and Nottinghamshire
Hertfordshire and West Essex
Bedfordshire Luton and Milton Keynes
Leicester Leicestershire and Rutland
Devon
Lincolnshire
Sussex
Humber and North Yorkshire
Gloucestershire
Cambridgeshire and Peterborough
Suffolk and North East Essex
Norfolk and Waveney
West Yorkshire
Derby and Derbyshire
Dorset
Northamptonshire
Bath and North East Somerset Swindon and Wiltshire
Mid and South Essex
Sub-Location


Total Appointments by Status and
1e8
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Count of Appointments
hcp_type
GP
Other Practice staff
Unknown
Attended
DNA
Unknown
Appointment Status

Scenario Analysis

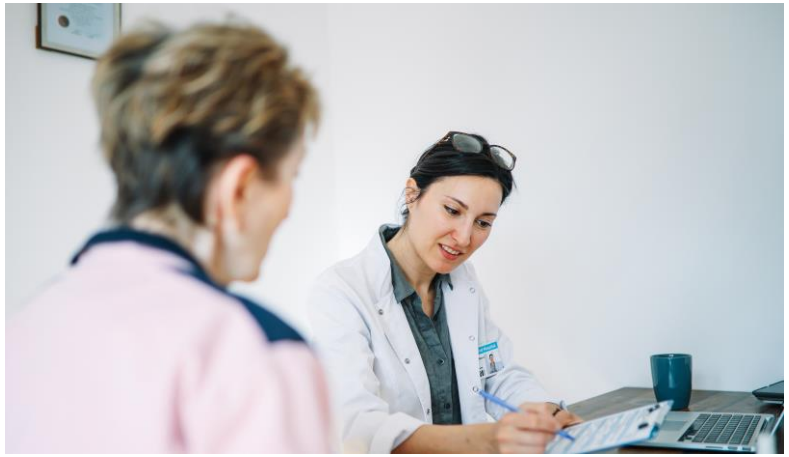

Weekday Daily Appointments: Original vs Adjusted (3% & 6% DNA)
Original
Adjusted (3% DNA)
Adjusted (6% DNA)
Frequency
0
10
20
30
40
50
0.8
1.0
1.2
1.4
1e6
Appointments per Day

Conclusions and recommendations

Key points

1. Missed appointments
2. Population demographics
3. Data consistency



Thank you

Peter Mackin